

Sarthak Bichhawa

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Summary

Senior Data & Applied Scientist and org-recognized AI Champion (9+ years) shipping production AI and ML at cloud scale—autonomous/multi-agent LLM systems, behavioral embeddings, and revenue-critical data platforms; recent impact includes ~\$45M/month influenced revenue recovered and multi-hour incident triage cut to minutes.

Experience

Microsoft, Redmond, WA | **Sr. Data & Applied Scientist**

Nov 2020 – Current

AI & Agentic Systems:

- Cut customer-health time-to-insight from **days to <2 minutes** by architecting an enterprise LLM agent for **Mission-Critical Customers (MCC)** that turns natural-language questions into governed **live Kusto/ADX queries** and returns PM-ready narratives, grounded on a curated catalog of 13 metric domains.
- Reduced multi-hour on-call incident investigations to **minutes** with an **autonomous, multi-agent incident-investigation agent**—a planner/executor LLM loop on Azure OpenAI with context compaction, confidence scoring, and run tracing (FastAPI on Azure); shipped two reusable **Model Context Protocol (MCP) tool servers** (one almost entirely AI-generated) serving governed query and telemetry tools to Copilot, VS Code, and internal agents.

Applied ML & Data-Quality Platform:

- Enabled fleet-scale similarity search, segmentation, and **embedding-drift anomaly detection across millions of accounts** by designing an **account-behavior embedding model** (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.
- Caught silent upstream data regressions—including an **~88% row-collapse**—before they reached dashboards or Finance by building a self-service **data-quality monitoring service** (Python/FastAPI, anomaly detection, 76 tests, Azure Container Apps) adopted across teams, plus an hourly table-health profiling layer over 47 datasets.

Partner Revenue & Business Reporting:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (at <1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component partner-attribution pipeline with multi-partner credit replacing last-writer-wins.
- Collapsed monthly business-review reporting from a multi-day manual process to a governed **~5-minute pipeline (>95% effort reduction)**, now in regular leadership use; shipped Files/Disk/Blob business-insights dashboards on a curated dataset layer.

Reliability & Customer Health:

- Helped prevent **10+ Sev1/Sev2 incidents** with adaptive pre-release stress-testing for Azure Storage (multi-armed bandits; 60% fewer false positives); drove the Mission-Critical Customers health program across 8+ teams (~50% fewer aging issues, ~16% faster time-to-mitigate) and shipped 15+ Azure Advisor recommendations at scale.

Leadership & Recognition:

- As an org-recognized **AI Champion**, drove AI adoption across the team—co-led a 90-minute internal **AI Day-of-Learning**, mentored engineers (reviewed 17 PRs, onboarded a new hire), and led cross-org incident response (4-team Sev-2).

Walmart Labs, Bentonville, AR | **Senior Data Scientist**

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift across 50 stores** with an end-to-end planogram optimization system (planning cycles weeks→hours) plus product association/substitution analytics for assortment and replenishment.
- Led 4 DS/analysts + 2 interns to productionize a repeatable item-level placement workflow; earned a Spot Award.

Integral Ad Science, New York, NY | **Data Scientist Intern**

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing ~1M URLs/day across 846 categories (88% accuracy).

Mu Sigma Business Solutions, Bengaluru, India | **Data Scientist**

Oct 2015 – July 2017

- Estimated customer lifetime value (CLTV) using probabilistic/regression models; improved competitive reporting via fuzzy product-ID matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | **MS-Business Analytics**

Aug 2017 – Dec 2018

SGSITS, Indore, India | **B.E. in Computer Engineering**

Aug 2011 – July 2015

Skills

- **AI / ML:** Generative & agentic AI, autonomous/multi-agent systems & agent orchestration (planner/executor, tool-calling, RAG/grounding), Model Context Protocol (MCP) servers & clients, Azure OpenAI, LLM observability/tracing, LLM/agent evaluation, vector embeddings & autoencoders, FAISS, anomaly detection, multi-armed bandits, production ML
- **Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), Power BI, FastAPI, Azure DevOps, PySpark, R, Git
- **Platforms & Functional Areas:** Azure (App Service, Container Apps, Data Factory), large-scale telemetry & data-quality pipelines, partner/revenue attribution & business reporting, customer & retail analytics